

Ticket #006A: Missing Default Gateway

Symptoms: Server is reachable on the internal network but unable to communicate with external endpoints. Monitoring alerts indicate loss of connectivity to upstream services.

Resolution:

1. Confirming the failure from the affected server.

```
[jay@mainserver:~$ ping -c 4 8.8.8.8  
ping: connect: Network is unreachable
```

Analysis: Network connectivity failure confirmed.

2. Verifying if the router is still reachable from the affected server.

```
[jay@mainserver:~$ ping -c 3 192.168.1.1  
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.  
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=7.25 ms  
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=2.91 ms  
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=6.74 ms  
  
--- 192.168.1.1 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2003ms  
rtt min/avg/max/mdev = 2.914/5.636/7.253/1.936 ms
```

Analysis: The router is still reachable from the affected server.

3. Confirming that the affected server is reachable from other clients on the same local network.

```
[alejandrorabadan@T900s-MacBook-Air ~ % ping -c 4 192.168.1.228  
PING 192.168.1.228 (192.168.1.228): 56 data bytes  
64 bytes from 192.168.1.228: icmp_seq=0 ttl=64 time=1.578 ms  
64 bytes from 192.168.1.228: icmp_seq=1 ttl=64 time=0.942 ms  
64 bytes from 192.168.1.228: icmp_seq=2 ttl=64 time=0.817 ms  
64 bytes from 192.168.1.228: icmp_seq=3 ttl=64 time=0.819 ms  
  
--- 192.168.1.228 ping statistics ---  
4 packets transmitted, 4 packets received, 0.0% packet loss
```

Analysis: Network connectivity from the local network successfully confirmed.

4. Verifying the server's default route.

```
[jay@mainserver:~$ ip route
192.168.1.0/24 dev enp0s1 proto kernel scope link src 192.168.1.228 metric 100
192.168.1.1 dev enp0s1 proto dhcp scope link src 192.168.1.228 metric 100
```

Analysis: There is no default gateway configured on the server. When the server attempts to reach external networks, packets destined for non-local subnets are forwarded to the default gateway. Since no default route exists in the routing table, these packets are dropped by the kernel.

5. Adding the default gateway to the server's routing table, pointing it to the network gateway via interface enp0s1, and testing internet connectivity.

```
[jay@mainserver:~$ sudo ip route add default via 192.168.1.1 dev enp0s1
[sudo] password for jay:
[jay@mainserver:~$ ip route
default via 192.168.1.1 dev enp0s1
192.168.1.0/24 dev enp0s1 proto kernel scope link src 192.168.1.228 metric 100
192.168.1.1 dev enp0s1 proto dhcp scope link src 192.168.1.228 metric 100
[jay@mainserver:~$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=31.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=17.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=28.1 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 17.941/25.986/31.897/5.894 ms
```

Analysis: After manually adding the correct gateway, internet connectivity is restored, confirming the routing configuration as the root issue.

6. Further investigation is required to determine why the DHCP process did not automatically assign the default gateway.
7. See T6B: Missing Default Gateway DHCP Investigation.